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## UNIT 18 FISCAL POLICY\*

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### 18.0 OBJECTIVES

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After going through the unit you will be in a position to

- explain the concepts of fiscal policy, government budget, and budget balance;
- distinguish between Discretionary and Non-Discretionary fiscal policies;
- identify the role of these policies in reducing output fluctuations; and
- explain the issues related to the implementation of fiscal policy.

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### 18.1 INTRODUCTION

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You may be aware by now that the economy does not always work smoothly. There often occur fluctuations in the level of economic activity. At times the economy finds itself in the grip of recession when levels of income, output and employment are below their full employment levels. At other times the economy is overheated, which means occurrence of rising prices and high inflation. The classical economists believed that an automatic mechanism works to restore stability in the economy; recession would cure itself and inflation will be automatically controlled. However, the empirical evidence during the 1930s when severe depression took place in the western capitalist economies shows that no such automatic mechanism works to bring about stability in the economy. In view of this Keynes argued for intervention by the government to cure depression and inflation by adopting appropriate tools for macro-economic policy. The two important tools of macroeconomic policy are fiscal policy and monetary policy. We have already dealt with monetary policy in the previous two units.

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In the present unit we will dwell upon the role of fiscal policy for stabilizing the economy.

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## 18.2 GOALS OF FISCAL POLICY

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Fiscal policy basically is demand management policy in the sense that it influences aggregate demand through changes in government expenditure. Taxes, transfer payments, and government expenditures on goods and services are the tools of fiscal policy. The main objective of fiscal policy is to stabilize output by managing aggregate demand, keeping output close to potential output, and reducing the size and duration of business cycle fluctuations. This requires changes in the government's expenditure plans and tax policy to offset changes in autonomous consumption, investment and exports that would otherwise push the economy away from equilibrium at potential output.

Fiscal Policy tools, taxes, transfers and government expenditure, influence aggregate demand through disposable income and consumption expenditure, and directly through government expenditures. The net tax rate also affects the size of the expenditure multiplier (also called Keynesian multiplier or spending multiplier). Thus, changes in tax rates, transfer rates, and government expenditures shift the AD curve.

At the time of recession the Government increases its expenditure or cuts down taxes or adopts a combination of both. On the other hand, to control inflation the Government cuts down its expenditure or raises taxes. In other words, to cure recession, expansionary fiscal policy and to control inflation, contractionary fiscal policy is adopted. Adoption of such expansionary fiscal policy will cause deficit in the government's budget and contractionary fiscal policy will cause a surplus in the government's budget. As the government implements fiscal policy through its budget, let us first understand the budget balance.

The government budget describes what goods and services the government will buy during the coming year, what transfer payments it will make, and how it will pay for them. When revenues and spending are equal, the budget is balanced. When revenues exceed spending, there is a budget surplus. When revenues fall short of spending, there is a budget deficit. The government budget balance is the difference between revenues and expenditures.

$$\text{Government Budget Balance (BB)} = NT - G = (tY - G)$$

Where  $G$  = Government Purchases,  $NT$  = net Tax Revenue and,  $t$  = tax rate

The budget balance, whether deficit, surplus or zero, is determined by three things:

1. The *net tax rate*  $t$  set by the government;
2. The level of *expenditure*  $G$  set by the government; and
3. The level of *output*  $Y$  determined by AE and AD

In Fig. 18.1, panels (a) and (b), we present the budget function and the budget balance respectively. In Fig 18.1 panel (a), the budget is balanced (i.e.,  $G=NT$ ) at the aggregate income level  $Y_1$ . The budget has a deficit for income level less than

$Y_1$  (i.e.,  $Y < Y_1$ ) and a surplus for income level more than  $Y_1$  (i.e.,  $Y > Y_1$ ). In Fig. 18.1 panel (b), the budget balance (BB) is determined by the level of GDP. A budget function shows us the different budget balances for one fiscal policy program at different levels of national income. Finally, notice that because a budget function describes one fiscal plan, any change in the fiscal plan will change the  $BB$  line to show a new budget function. We observe that  $BB=0$  (i.e.,  $NT=G$ ), for income level,  $Y_1$ . Correspondingly,  $BB>0$  for income levels greater than  $Y_1$  and  $BB<0$  for income levels less than,  $Y_1$ .

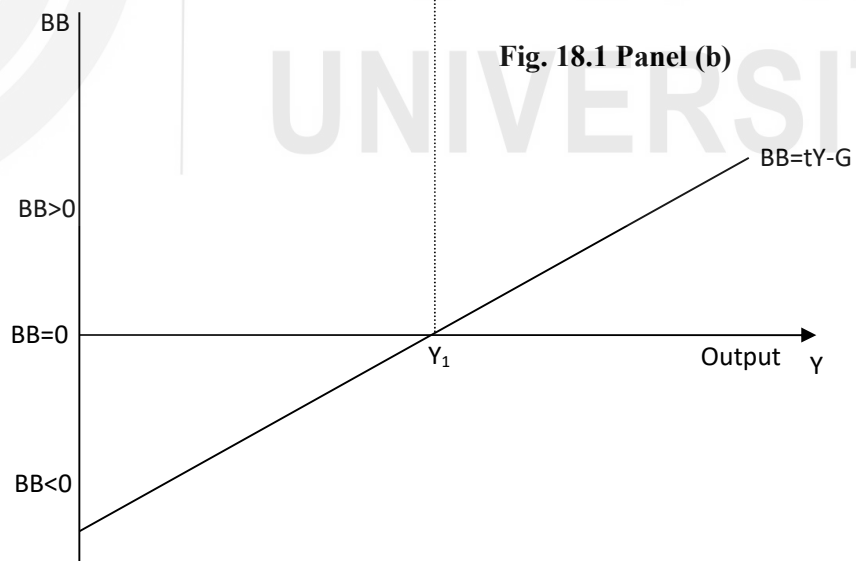
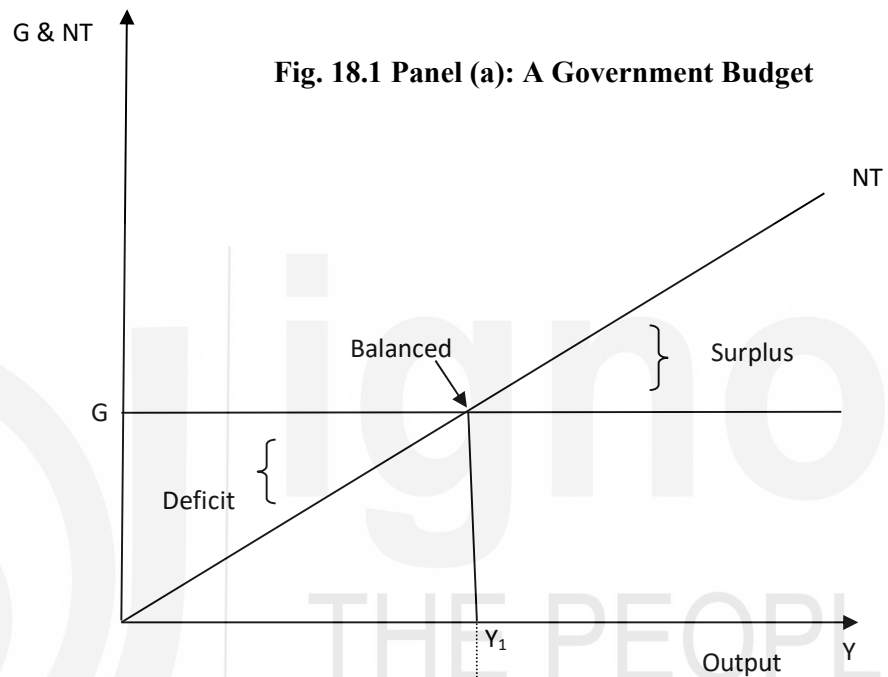


Fig. 18.1 shows the government budget is in balance at income level,  $Y_1$ . It is in surplus to the right of  $Y_1$  ( $Y > Y_1$ ,) and in deficit to the left of  $Y_1$  ( $Y < Y_1$ ,).

- 1) What is meant by fiscal policy?

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- 2) Explain the type of fiscal policies governments use to control recession and inflation?

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### 18.3 DISCRETIONARY FISCAL POLICY

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Fiscal policy is of two kinds, viz., i) Discretionary fiscal policy, and ii) Non-Discretionary fiscal policy of automatic stabilizers. By discretionary fiscal policy we mean deliberate change in government expenditure and taxes to influence the level of national output and prices. These are non-mandatory changes in taxation, spending, or other fiscal activities by a government in response to economic events or changes in economic conditions. Discretionary fiscal policy implies government actions above and beyond existing fiscal policies, and often occurs in periods of recession or economic turbulence. Discretionary fiscal policy relates to demand management that uses government spending and taxation policy to influence aggregate demand. The government uses an expansionary fiscal policy to correct a recessionary gap and a contractionary fiscal policy to correct an inflationary gap.

#### 18.3.1 Discretionary Fiscal Policy to cure Recession

The recession in an economy occurs when aggregate demand decreases below the full employment level due to a fall in investment, consumption or net exports. This fall in aggregate demand below the full employment level creates a deflationary or recessionary gap. It is the task of fiscal policy to close this gap by increasing government expenditure or reducing the taxes. We discuss below policy instruments used to cure recession.

### a) Increase in Government Expenditure

For a discretionary fiscal policy to cure depression, the increase in government expenditure is an important tool. Government may increase expenditure by starting public works. For undertaking all these public works, government buys various types of goods and materials and employs workers. The effect of this increase in expenditure is both direct and indirect. The direct effect is the increase in incomes of those who sell materials and supply labour for these projects. The output of these public works also goes up together with the increase in incomes. Not only that, Keynes showed that increase in government expenditure also has an indirect effect in the form of the working of a multiplier (see Block 1, Unit 2). Those who get more income spend them further on consumer goods depending on their marginal propensity to consume. The impact of increase in government expenditure in a recessionary condition is illustrated in Fig 18.2. The aggregate expenditure is given by the curve AE while the  $45^\circ$  line depicts the equality between aggregate output (Y) and AE. Suppose the initial aggregate expenditure,  $AE_0$ , intersect the  $45^\circ$  line at point  $E_1$  and thereby determine  $Y_0$  as the level of output. As this level of equilibrium output is below the full employment level ( $Y_f$ ), a deflationary gap of  $Y_0Y_f$  is created. We observe from Fig. 18.2 that, the deflationary gap can be eliminated if the government increases its expenditure by  $E_1H$ . The aggregate expenditure curve will shift upwards to  $AE_1$  and as a result the equilibrium level of income will increase to the full employment level,  $Y_f$ . Note that the increase in output by  $Y_0Y_f$  is more than the increase in government expenditure  $\Delta G = E_1H$ , due to multiplier effect.

**Fig. 18.2: Increase in Government Expenditure**

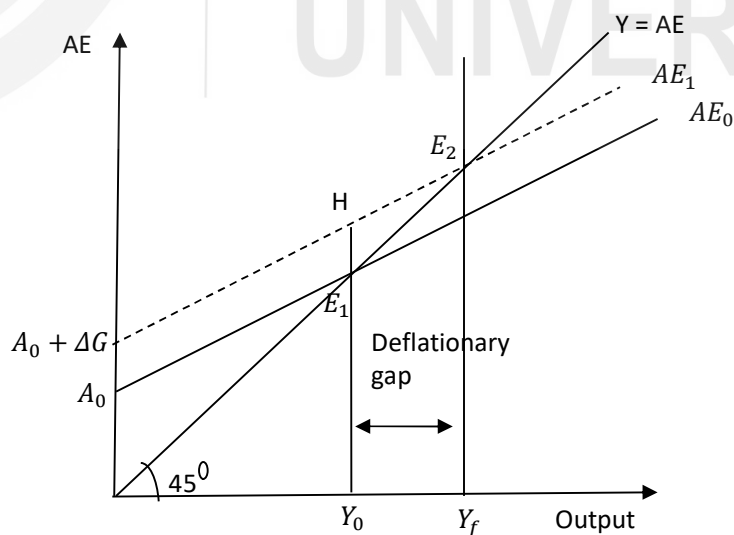


Fig.18.2 shows that an increase in Government expenditure increases Y and closes the deflationary gap.

## b) Reduction in Taxes

An alternative fiscal measure to overcome recession and to achieve expansion in output and employment is reduction in taxes. The reduction in taxes increases the disposable income of the society and causes the increase in the consumption spending of the people. Thus reduction in taxes will cause an upward shift in the consumption function. If along with the reduction in taxes, the government expenditure is kept unchanged, aggregate expenditure curve will shift upward due to rise in consumption function curve. This will have an expansionary effect and the economy will be lifted out of recession, and national income and employment will increase. Fig. 18.3 shows the effect of a reduction in taxes on addressing the problem of recessionary gap. Aggregate demand curve  $AE_0$ , intersect the  $45^\circ$  line at point  $E_1$ , with  $Y_0$ , as the equilibrium level of income. A recessionary gap of  $Y_0 Y_f$  exists as this level of output falls short of the full employment level,  $Y_f$ . A fall in the tax rate, increases the slope of the aggregate expenditure curve (by increasing the value of the multiplier) and shifts the aggregate expenditure curve outward to  $AE_1$ , which intersects with the  $Y=AE$  line at point  $E_2$ . As the level of equilibrium income now becomes equal to the full employment level of income,  $Y_f$ , the recessionary gap disappears. Thus, a reduction in taxes will increase consumption expenditure  $C$ , and shifts the aggregate expenditure curve upward and will result in, through the working of the multiplier, a higher level of equilibrium level of income.

**Fig 18.3: Increase in Net Tax Rate**

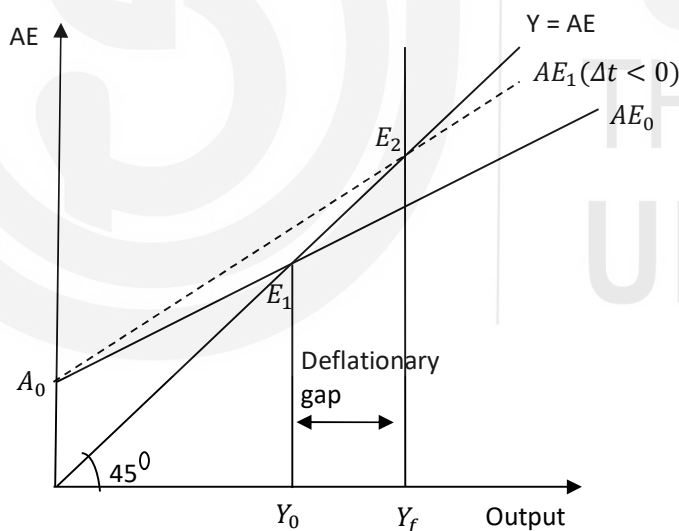


Fig 18.3 shows that a decrease in net tax rate increases spending and the multiplier to close the recessionary gap.

### 18.3.2 Discretionary Fiscal Policy to Control Inflation

A situation of excess demand in the economy can arise because of three reasons: i) increase in household demand due to large increase in consumption, ii) increase in demand by firms due to increase in investment, and iii) increase in government expenditure. As aggregate expenditure increases beyond what the

economy can potentially produce by fully employing its given resources, the situation of excess demand builds inflationary pressures in the economy. An inflationary situation can also arise if too large an increase in money supply in the economy occurs. In these circumstances, an inflationary gap occurs which tend to bring about rise in prices. Under such circumstances anti-cyclical fiscal policy calls for reduction in aggregate expenditure. Thus fiscal policy measures to control inflation are (a) reducing government expenditure (b) increasing taxes.

#### a) Reducing Government Expenditure

Fig. 18.4 shows how the reduction in government expenditure will help checking inflation. You can see from the figure that an aggregate demand,  $AE_0$  intersects the  $45^\circ$  line at point,  $E_1$ , and determines equilibrium level of income at  $Y_0 > Y_f$ , (the full employment level). As a result, an inflationary gap equal to  $Y_0 - Y_f$  emerges. A reduction in government expenditure equal to  $E_1H$ , reduces aggregate expenditure and shifts the curve down to  $AE_1$ , which will restore the equilibrium at the full employment level,  $Y_f$ . The reduction in Government expenditure equal to  $E_1H$  through the operation of multiplier will result in a multiple decline in the level of national income or output. Fig. 18.4 shows that the decrease in government expenditure by  $E_1H$  has led to a much bigger decline in output by  $Y_0 - Y_f$ .

**Fig. 18.4: Decrease in Government Expenditure**

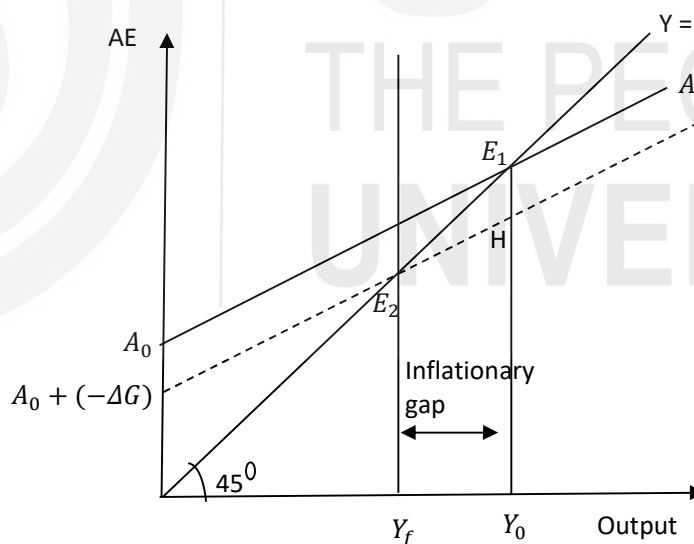


Fig. 18.4 shows that a decrease in Government expenditure decreases  $Y$  and closes the inflationary gap.

#### b) Increase in Taxes

In addition to the reduction in Government expenditure, the government can increase taxes to reduce aggregate demand. The hike in taxes reduces the disposable income of the people and thereby forces them to reduce their

consumption demand. In Fig. 18.5, the inflationary gap of  $Y_0 Y_f$  is closed by an increase in the net tax rate which reduces the slope of AE and shifts the aggregate expenditure downward to  $AE_1$ . This downward shift of the aggregate expenditure curve moves the economy back to the full employment equilibrium at,  $Y_f$ .

**Fig. 18.5: Increase in Net Tax Rate**

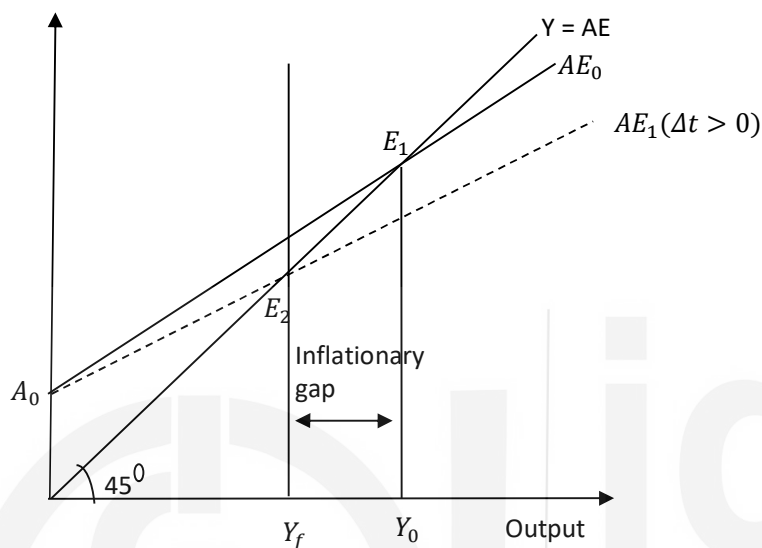


Fig. 18.5 shows that an increase in net tax rate reduces spending and the multiplier to close the inflationary gap.

### 18.3.3 Time Lags in Implementation of the Discretionary Fiscal Policy

Although the government has long conducted monetary and fiscal policy, the view that it should use these policy instruments to try to stabilize the economy is more recent. To many economists the case for active government policy is clear and simple. Recessions are periods of high unemployment, low incomes, and increased economic hardship. The model of aggregate demand and aggregate supply shows how shocks to the economy can cause recessions. It also shows how monetary and fiscal policy can prevent (or at least soften) recessions by responding to these shocks. These economists consider it wasteful not to use these policy instruments to stabilize the economy. Other economists are critical of the government's attempts to stabilize the economy. These critics argue that the government should take a hands-off approach to macroeconomic policy. Why do these critics want the government to refrain from using monetary and fiscal policy for economic stabilization? To find out, let's consider some of their arguments.



Stabilization goals are not easy to achieve. The existence of various kinds of time lags, or delays in the response of the economy to stabilization policies, can make the economy difficult to control. Indeed, the problem for policymakers is even more difficult, because the lengths of the lags are hard to predict. Economists distinguish between two lags that are relevant for the conduct of stabilization policy: the inside lag and the outside lag. The inside lag is the time between a shock to the economy and the policy action responding to that shock. This lag arises because it takes time for policymakers first to recognize that a shock has occurred and then to put appropriate policies into effect. The outside lag or the response lag is the time between a policy action and its influence on the economy. This lag arises because policies do not immediately influence spending, income, and employment. The inside lag can be further classified into the recognition lags and the implementation lags.

**a) Recognition Lag**

It takes time for policy makers to recognize a boom or a slump. Many important data—those from the national income and product accounts, for example—are available only quarterly. It usually takes several weeks to compile and prepare even the preliminary estimates for these figures. If the economy goes into a slump on January 1, the recession may not be detected until the data for the first quarter are available at the end of April.

**b) Implementation Lag**

The problems that lags pose for stabilization policy do not end once economists and policy makers recognize that the economy is in a boom or a slump. Even if everyone knows that the economy needs to be stimulated or reined in, it takes time to put the desired policy into effect, especially for actions that involve fiscal policy. Implementation lags result.

**c) Response Lag**

Even after a macroeconomic problem has been recognized and the appropriate policies to correct it have been implemented, there are response lags – lags that occur because of the operation of the economy itself. For example, it takes time for the multiplier to reach its full value. The delay in the multiplier process occurs because neither individuals nor firms revise their spending plans instantaneously. Until they can make those revisions, extra government spending does not stimulate extra private spending.

In practice, it takes about a year for a change in taxes or in government spending to have its full effect on the economy. This response lag means that if we

increase spending to counteract a recession today, the full effects will not be felt for 12 months. By that time, the state of the economy might be very different.

The long and variable lags associated with fiscal policy certainly make stabilizing the economy more difficult. Advocates of passive policy argue that, because of these lags, successful stabilization policy is almost impossible. Indeed, attempts to stabilize the economy can be destabilizing. Suppose that the economy's condition changes between the beginning of a policy action and its impact on the economy. In this case, active policy may end up stimulating the economy when it is heating up or depressing the economy when it is cooling off. Advocates of active policy admit that such lags do require policymakers to be cautious. But, they argue, these lags do not necessarily mean that policy should be completely passive, especially in the face of a severe and protracted economic downturn.

### Check Your Progress 2

- 1) Show diagrammatically how a recession can be controlled through the use of discretionary fiscal policy.

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## 18.4 NON-DISCRETIONARY FISCAL POLICY

Non-discretionary fiscal policy of automatic stabilizers is a built-in tax or expenditure mechanism that automatically increases aggregate demand when recession occurs and reduces aggregate demand when there is inflation in the economy without any special deliberate actions on the part of the government.

These policies, called automatic stabilizers, are designed to reduce the lags associated with stabilization policy. Automatic stabilizers are policies that stimulate or depress the economy when necessary without any deliberate policy change. In the non-discretionary fiscal policy, the tax structure and expenditure pattern are so designed that taxes and government spending vary automatically in appropriate direction with the changes in national income. That is, these taxes and expenditure pattern without any special deliberate action by the government